Quadratic expressions

Substitute into squared terms, then follow the order of operations.

Name: _____ Suggested time: 10 minutes

Show your working. Use brackets when substituting negative values.

01 Given x=5

Evaluate: x^2+2x

02 Given x=-3

Evaluate: x^2+3x+5

03 Given x=-4

Evaluate: x^2+3x-1

04 Given x=-1

Evaluate: x^2+5x-8

05 Given x=3

Evaluate: x^2+4x-7

06 Given x=-4

Evaluate: x^2+2x+6

Answer sheet

Quadratic expressions | 6 questions

Keep this section separate from the question sheet. Full working follows on the next pages.

01 35

02 5

03 3

04 -12

05 14

06 14

Worked solutions

Quadratic expressions

01 Given x=5

x^2+2x

Substitute x=5.

$(5)^2 +2\times 5$

$25 +10$

35

02 Given x=-3

x^2+3x+5

Substitute x=-3.

$(-3)^2 +3\times (-3) +5$

$9 -9 +5$

5

03 Given x=-4

x^2+3x-1

Substitute x=-4.

$(-4)^2 +3\times (-4) -1$

$16 -12 -1$

3

04 Given x=-1

x^2+5x-8

Substitute x=-1.

$(-1)^2 +5\times (-1) -8$

$1 -5 -8$

-12

Worked solutions

Quadratic expressions

05 Given x=3

x^2+4x-7

Substitute x=3.

$(3)^2 +4\times 3 -7$

$9 +12 -7$

14

06 Given x=-4

x^2+2x+6

Substitute x=-4.

$(-4)^2 +2\times (-4) +6$

$16 -8 +6$

14